RESEARCH STATEMENT

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I study how deeper integration into the world economy changes the decisions people make in countries more vulnerable to globalization. Low- and middle-income countries have leaned on foreign capital and market access to raise growth and productivity, and the aggregate returns to that strategy are well documented. Far less studied, at least in low- and middle-income settings, is how integration reshapes individual choices. My work studies those channels. Costa Rica is my primary setting because it supplies the variation these questions require: free trade zone policy that concentrated foreign direct investments in particular industries, a national referendum on CAFTA-DR that made trade agreements a political process, and administrative records detailed enough to observe individual college major choices. Familiarity with the institutional context and the language is what lets me use them well.

GLOBAL FIRMS, LOCAL STUDENTS: MULTINATIONAL PRESENCE SHAPES COLLEGE MAJOR *(Job Market Paper)*

Students respond to the industrial composition of foreign investment around them when choosing what to study. Using administrative application records from a large public university in Costa Rica covering 2010 to 2019 and 230,034 applicants, I estimate how multinational presence in an industry shifts the probability that a prospective student selects the field of study serving it. I construct a measure of local exposure that weights free trade zone firms by their distance to each applicant's district of residence and by how intensively each industry hires from each field. Identification comes from changes in nearby firm presence within districts over time, holding fixed any district-specific or year-specific determinant of field choice; standard errors are clustered at the district level, since exposure is assigned at district-year rather than to individual students. Estimating the choice problem as a conditional logit and reporting average marginal effects, I find that a one standard deviation increase in within-district manufacturing presence raises the probability of choosing a major within STEM and Applied Sciences by 2.0 percentage points — five percent of the baseline rate — drawn four-fifths from Social Sciences and Professional Studies (-1.6 p.p.) and one-fifth from Arts, Writing and Culture (-0.4 p.p.). Professional, scientific and technical activities move students in the opposite direction, toward business and professional programs. Because exposure to different industries is highly correlated across space, this divergence in signs cannot be produced by a common factor such as local economic development, and is consistent with a selective response to industry-specific labor demand rather than a general shift in student preferences.

Status: Working Paper. The most recent available draft can be found on my website.

DOES THE GLOBAL SOUTH IMPORT IDEOLOGY SHIFTS? EVIDENCE FROM COSTA RICA

Rising trade volume between Costa Rica and the United States is associated with a relative rightward shift in political ideology. I combine political party manifesto data with legislative election returns to construct a measure of ideology at the canton level, then use a shift-share design with instrumented bilateral trade volume to estimate canton-level import exposure. More exposed cantons shift conservative, by 1.83 standard deviations in the political lean index. I additionally use canton-level voting on the CAFTA-DR referendum to proxy for attitudes toward free trade, and find that a

favorable prior attitude attenuates the conservative shift. This suggests the response is mediated by how trade is understood locally, not only by exposure itself.

Status: Working Paper. The most recent available draft can be found on my website

MULTINATIONAL ENTRY AND SKILL COMPOSITION - A THEORETICAL MODEL

The empirical chapter establishes that multinational entry changes what students choose to study. It however cannot say what happens to aggregate skill composition of the economy once those individual responses feed back into wages and into multinational firms own location and hiring decisions. This chapter builds a general equilibrium model in which multinational and domestic firms compete for workers while students choose skill acquisition anticipating the returns those firms offer, so that entry and specialization are jointly determined rather than sequential.

ONGOING AND FUTURE WORK

ENVIRONMENTAL POLICY AND FDI TYPE w/ ANCA CRISTEA AND BRUCE BLONIGEN

Enivronmental regulation may not deter foreign investment so much as redirect the form it takes. Greenfield projects and cross-border acquisitions face asymmetric regulatory exposure: a new plant must satisfy current standards in full, while an acquired plant is often granfathered in, with compliance costs capitalized into th epurchase price instead. If that asymmetry binds, stringency should reallocate FDI across modes even where it leaves the total unchanged. I test this using the OECD Environmental Policy Stringency Index, which rose sharply across member countries between 1990 and 2020, together with the growth in environmental provisions written directly into preferential trade agreements. The setting offers substantial cross-country variation in how aggressively host nations regulate while competing for the same foreign capital. Distinguishing deterrence from reallocation matters both for the theory of multinational firm organization and for policymakers who assume stringency and investment trade off one for one.

MEXICO-US MIGRATION AS A DIFFUSION PROCESS

I model migration from Mexico to the United States as a single continuous process spanning emigration, border crossing, and settlement, rather than as three separately estimated stages. Adapting a convection-difussion equation, I treat migrant flows as a density field evolving over space and time: where flows spread, drift toward opportunity, attrit under enforcement, and accumulate in established ethnic enclaves. The framework joins two spatial regimes: a corridor capturing pre-border dynamics within Mexico and a two-dimensional diffusion system capturing settlement across the United States. These are both linked at the border by an enforcement-dependent permeability condition. I estimate the parameter vector, including the wage-gap elasticity, the enforcement elasticity of crossing probability, and a network elasticity governing interior diffusivity, by using simulated method of moments against county-year density moments from the American Community Survey.

RESEARCH AND TEACHING FIT

Additionally, my future work aims to use publicly available datasets such as the OECD Environmental Policy Stringency Index for the environmental policy project and the American Community Survey for the language acquisition project. At this current stage, none of this work requires overwhelming external funding, restricted-access agreements, laboratory infrastructure, or fieldwork. This sets up all of my future projects to involve any student at any stage of research skills to join and learn from the process.

I am happy to involve students in this work and equally glad to advise projects of their own. What I most want to offer to students is exposure to the part of research that coursework rarely reaches: obtaining the data, cleaning it, and deciding how to turn a concept into something measurable. Students are faced with regression outputs long before they encounter the judgment calls that produce it. Working through those decisions on a live project, whether mine or theirs, is where a student stops applying methods and starts doing economic research.